

Adilet Shynggys

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RESEARCH INTERESTS

Heat transfer, thermodynamics, and two-phase flow for cleaner, more efficient, and safer power generation, particularly nuclear energy: how microscale surface geometry, texture, and wettability govern the supply of liquid to heated surfaces, rewetting, boiling, and the boiling crisis, and how those effects carry to reactor thermal-hydraulics.

EDUCATION

Nazarbayev University — B.Eng., Mechanical and Aerospace Engineering

Expected June 2027

Astana, Kazakhstan

- GPA **3.86 / 4.00**; **ranked 1st** in cohort; Dean's List, four consecutive semesters.
- Coursework: Engineering Thermodynamics, Fluid Mechanics, Heat and Mass Transfer (in progress), Structural Mechanics, Materials and Manufacturing, Numerical Methods, Engineering Dynamics I–II.

The University of Hong Kong — Visiting Exchange Student, Mechanical Engineering

Spring 2026

Hong Kong SAR

- Energy Conversion Systems, Design and Manufacture, Additive Manufacturing; collaborated with Prof. Wei Yan on curriculum design for an undergraduate course.

RESEARCH AND DESIGN EXPERIENCE

Capillary-Driven Surface Engineering

Aug 2026 – Present

Dept. of Mechanical and Aerospace Engineering, Nazarbayev University

Advisor: Prof. Amgad Salama

- Studying how microchannel taper geometry moves liquid with no pump: the taper creates a Laplace-pressure asymmetry that should drive wetting liquids toward the narrow end and non-wetting liquids toward the wide end.
- With the group, designed the inverted photomask and completed the full clean-room photolithography process, producing a first successful sample of tapered SU-8 microchannels on silicon.
- Next: transport experiments across taper geometries and droplet- and slug-velocity measurement, toward understanding how geometry speeds the rewetting of heated surfaces.

Senior Capstone (Engineering Design): VVER-1200 Reactor Core

Aug 2026 – Present

Nazarbayev University • Team of four

Advisor: Prof. Amgad Salama

- Converted the required electrical output to 3,200 MWth of core thermal power via cycle efficiency, and used a target burnup near 50 GWd/tHM to size fuel pellets, rods, and bundles into a hexagonal core configuration.
- Now entering thermal-hydraulic analysis: coolant flow and temperature rise, critical heat flux, and margins to subcooled boiling and DNB, feeding a reduced-order model to be validated against published reference data.

Triboelectric Nanogenerators for Wearable Sensing

Jan – Dec 2025

Energy Conversion Laboratory, Nazarbayev University

Supervisor: Dr. Shabnam Yavari

- Fabricated triboelectric nanogenerators with the group by electrospinning and silicone casting, raising output power density by 15–20%.
- Ran device characterization across humidity and cyclic durability, and built the electrical workflow behind it: load sweeps, data acquisition, and analysis in OriginPro and MATLAB.
- Results contributed to a co-authored article in *Small* and a first-author poster awarded **Best Research Poster**.

PUBLICATIONS AND PRESENTATIONS

- [1] S. Faramarzi, S. Yavari, U. Ospanova, **A. Shynggys**, et al., “Polarity-Engineered, Tribo-Positive Imidazolium-Mediated Crosslinked 6FDA-Durene Films for Triboelectric Nanogenerator,” *Small*, 2026, e74486. doi:10.1002/smll.74486
- [2] **A. Shynggys**, S. Yavari, G. Kalimuldina, “Green-Synthesized Ag Nanoparticles Incorporated into High-Performance Triboelectric Wearable Sensors.” Poster, Belt and Road Initiative Conference, Astana, 2025 (**Best Research Poster**); abstract, INESS-2025, Astana, Aug. 2025.

TEACHING, LEADERSHIP, AND SERVICE

Student Advisory Body, School of Engineering — elected undergraduate representative 2026 – Present

- One of two students elected for the major, one undergraduate and one graduate, carrying student concerns and recommendations directly to the Dean and Vice Dean.

Physics Tutor (volunteer), Nazarbayev University — tutored **20+** first-year students 2023 – 2024

Cohort Representative, Mechanical and Aerospace Engineering 2024 – 2026

- Ran **10+** exam-preparation nights for the cohort.
- Raised student requests with the dean, leading to a rescheduled exam and more exchange opportunities.

Community Service 2025 – 2026

- Sponsorship team, charity trail run supporting MamaPro, a nonprofit providing business education to single mothers: **150** runners, **600,000 KZT** (~\$1,300) raised.
- Volunteer, Mereke Charity Foundation (retirement-home visits, winter clothing drive for children from vulnerable families) and HKU Muslim Society (organizing iftars during Ramadan).

OTHER DESIGN PROJECTS

- *Multi-stage gearbox* (University of Hong Kong): sized shafts by the ASME bending–torsion criterion and selected gears, keyways, and bearings for the required fatigue life.
- *Interactive art installation* (HKU, with Prof. Homayoon Kazerooni, UC Berkeley): modeled petal forms in SolidWorks, checked collisions and stability, and 3D-printed each piece.

INDUSTRY EXPERIENCE

Product Support Engineering Intern, Wabtec Corporation, Kazakhstan June – July 2025

Internship awarded as the Mechaton 1st-place prize

- Analyzed traction-motor thermal performance and supported root-cause review of temperature excursions.
- Audited shop-floor manufacturing against drawings, reconciling GOST with ASME, and traced an insulation inconsistency to different export-market climates.

AWARDS AND HONORS

Presidential Scholarship of the Republic of Kazakhstan (2026–27) | Yessenov Foundation Scholarship, 20 of 500+ applicants (2026–27) | NU President’s Scholarship (2024–25) | Best Research Poster, Belt and Road Initiative Conference (2025) | 1st Place, Mechaton Engineering Case Competition (2025)

SKILLS

Computing: MATLAB, Python, Mathematica, OriginPro, L^AT_EX | **CAD:** SolidWorks (CSWA certified) | **Lab:** photolithography, electrospinning, silicone casting, 3D printing, device characterization | **Languages:** Kazakh (native), Russian (fluent), English (IELTS 8.0)